

◆ MZ UCA — AWS CLOUD PRACTITIONER BOOTCAMP**STUDENT ROADMAP**

Training • Labs • Proof of Work • Cloud Thinking

● WELCOME TO THE BOOTCAMP

Welcome to the:

◆ MZ UCA AWS CLOUD PRACTITIONER BOOTCAMP

This program is designed to help you:

- understand cloud computing properly,
- develop operational thinking,
- build real AWS skills,
- prepare for certification,
- and create visible proof of learning.

This is not a passive learning program.

You will:

- learn,
 - practice,
 - document,
 - analyze,
 - and build.
-

● PROGRAM DETAILS

Category	Information
Duration	4 Weeks
Schedule	Monday • Thursday • Saturday
Time	20:00 – 21:00
Extra Session	Additional Architecture Intensive During Week 3
Learning Style	Live Sessions + Labs + GitHub Documentation
Goal	Build cloud understanding + proof of work

● WHAT YOU MUST PREPARE

◆ AWS Account

Every student must:

- Create AWS Free Tier account
 - Enable MFA
 - Configure billing alerts
 - Create IAM administrator user
 - Avoid using the root account daily
-

◆ GitHub Repository

Repository Name

aws-cloud-practitioner-bootcamp

◆ Repository Structure

aws-cloud-practitioner-bootcamp/

```
├── README.md
├── Week-1-Foundations/
├── Week-1-Security/
├── Week-2-Compute/
├── Week-2-Storage/
├── Week-2-Databases/
├── Week-3-Networking/
├── Week-3-Monitoring/
├── Week-3-Architecture/
├── Week-4-Billing/
├── Week-4-Exam/
└── screenshots/
```

● WEEK 1 — CLOUD FOUNDATIONS + SECURITY

◆ SESSION 1 — MONDAY

Cloud Computing Foundations

You Will Learn

- What cloud computing is
- Why companies migrate to cloud

- Traditional IT vs cloud environments
 - Cloud service models
 - Public vs private vs hybrid cloud
 - AWS global infrastructure
-

Live Demonstration

- AWS Console walkthrough
 - Region selection
 - AWS navigation basics
-

GitHub Tasks

Folder

Week-1-Foundations/

Files

cloud-basics.md

cloud-models.md

aws-global-infrastructure.md

By the End of This Session

You should understand:

- how cloud works,
 - why AWS exists,
 - and how AWS infrastructure is organized globally.
-

◆ SESSION 2 — THURSDAY

Shared Responsibility + Governance

You Will Learn

- Shared Responsibility Model
 - Governance fundamentals
 - Operational risk awareness
 - Misconfiguration dangers
 - Root account risks
 - Compliance awareness
-

Risk Scenario Analysis

You will analyze:

- public S3 exposure,
 - leaked credentials,
 - weak IAM permissions,
 - excessive access risks.
-

GitHub Tasks

Folder

Week-1-Security/

Files

shared-responsibility.md
governance-basics.md
risk-scenarios.md

By the End of This Session

You should begin thinking:

- like a cloud operator,
 - like a governance-aware professional,
 - and like a security-conscious engineer.
-

◆ SESSION 3 — SATURDAY

Identity & Access Management (IAM)

You Will Learn

- IAM fundamentals
 - Users vs Groups vs Roles
 - Policies and permissions
 - Least privilege principle
 - MFA implementation
 - IAM best practices
-

Live Demonstration

- Create IAM user
- Enable MFA

- Configure permissions
 - Observe policy restrictions
-

GitHub Tasks

Files

```
iam-basics.md  
least-privilege.md  
security-best-practices.md
```

Screenshots Required

Upload into:

screenshots/

Required screenshots:

- IAM user creation
 - MFA enabled
 - IAM policies
-

By the End of This Session

You should be able to:

- secure AWS access,
 - manage permissions properly,
 - and understand access control.
-

🕒 WEEK 2 — CORE AWS SERVICES

◆ SESSION 4 — MONDAY

Compute Services

You Will Learn

- Amazon EC2
- Instance types
- Public vs private IPs
- Scaling basics
- AWS Lambda
- Serverless computing

Decision Scenarios

You will discuss:

- when EC2 is appropriate,
 - when Lambda is appropriate,
 - operational trade-offs,
 - and cost implications.
-

Live Demonstration

- Launch EC2 instance
 - Connect to instance
 - Observe instance states
-

GitHub Tasks

Folder

Week-2-Compute/

Files

ec2-overview.md

lambda-basics.md

compute-decision-guide.md

By the End of This Session

You should understand:

- compute strategy,
 - serverless concepts,
 - and workload decision-making.
-

◆ SESSION 5 — THURSDAY

Storage + Data Protection

You Will Learn

- Amazon S3
- Object storage concepts
- Storage classes

- Durability vs availability
 - Backup concepts
 - Lifecycle policies
-

Live Demonstration

- Create S3 bucket
 - Upload files
 - Configure permissions
 - Observe public access risks
-

GitHub Tasks

Folder

Week-2-Storage/

Files

s3-notes.md

storage-strategy.md

backup-concepts.md

By the End of This Session

You should understand:

- cloud storage architecture,
 - scalability benefits,
 - and data protection basics.
-

◆ SESSION 6 — SATURDAY

Databases + Cloud Migration

You Will Learn

- Amazon RDS
 - Amazon DynamoDB
 - Relational vs NoSQL databases
 - Migration fundamentals
 - Modernization concepts
-

Migration Discussions

You will analyze:

- why companies migrate,
 - legacy infrastructure challenges,
 - cloud scalability advantages,
 - modernization strategies.
-

GitHub Tasks

Folder

Week-2-Databases/

Files

database-comparison.md

migration-basics.md

modernization-notes.md

use-case-mapping.md

By the End of This Session

You should understand:

- database fundamentals,
 - migration reasoning,
 - and modernization thinking.
-

© WEEK 3 — NETWORKING + MONITORING + ARCHITECTURE

◆ SESSION 7 — MONDAY

Networking Fundamentals

You Will Learn

- Amazon VPC
 - CIDR basics
 - Public vs private subnets
 - Route tables
 - Internet Gateway
 - Network isolation
-

Live Demonstration

- VPC walkthrough
 - Traffic flow visualization
 - Subnet explanation
-

GitHub Tasks

Folder

Week-3-Networking/

Files

vpc-basics.md

subnet-design.md

networking-terminology.md

By the End of This Session

You should understand:

- AWS networking,
 - segmentation,
 - and cloud traffic flow.
-

◆ SESSION 8 — THURSDAY

Network Security

You Will Learn

- Security Groups
 - Network ACLs
 - Open port risks
 - Public exposure risks
 - Misconfiguration awareness
-

Live Demonstration

- Modify Security Group rules
 - Observe inbound access concepts
 - Analyze exposure scenarios
-

GitHub Tasks

Files

network-security.md
attack-scenarios.md
security-misconfigurations.md

By the End of This Session

You should understand:

- cloud network protection,
 - access control,
 - and security exposure risks.
-

◆ SESSION 9 — SATURDAY

Monitoring + Operational Visibility

You Will Learn

- Monitoring concepts
 - Logging concepts
 - Amazon CloudWatch
 - Metrics vs logs
 - Operational visibility
-

Live Demonstration

- CloudWatch walkthrough
 - Metrics observation
 - Monitoring dashboards
-

GitHub Tasks

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Week-3-Monitoring/

Files

monitoring-basics.md
cloudwatch-overview.md
operational-visibility.md

By the End of This Session

You should understand:

- monitoring fundamentals,
 - visibility concepts,
 - and operational awareness.
-

◆ EXTRA SESSION — WEEK 3

Architecture + High Availability + Resilience

Additional Session

Tuesday • 20:00–21:00

You Will Learn

- Basic cloud architecture
 - High availability
 - Fault tolerance
 - Multi-AZ concepts
 - Elastic Load Balancers
 - Scalability basics
 - Disaster recovery awareness
-

Architecture Scenario

You will design:

- resilient cloud environments using:
 - EC2
 - S3
 - IAM
 - VPC
 - Multi-AZ thinking
-

GitHub Tasks

Folder

Week-3-Architecture/

Files

architecture-diagrams.md

high-availability.md

By the End of This Session

You should begin thinking:

- like a systems designer,
 - like a cloud architect,
 - and like a resilience planner.
-

© WEEK 4 — GOVERNANCE + EXAM EXECUTION

◆ SESSION 10 — MONDAY

Billing + Cost Optimization

You Will Learn

- AWS pricing models
 - Cost drivers
 - Budget awareness
 - Cost optimization
 - Reserved vs On-Demand pricing
 - Financial governance basics
-

Live Demonstration

- AWS Pricing Calculator walkthrough
 - Cost estimation examples
-

GitHub Tasks

Folder

Week-4-Billing/

Files

pricing-models.md

cost-optimization.md

budgeting-basics.md

By the End of This Session

You should understand:

- cloud financial management,
 - budgeting,
 - and optimization principles.
-

◆ SESSION 11 — THURSDAY

Exam Strategy + Interview Preparation

You Will Learn

- Certification question techniques
 - Elimination strategies
 - Scenario interpretation
 - Cloud reasoning methods
 - Interview communication basics
-

Scenario Practice

You will answer:

- Why private subnets matter
 - Why least privilege matters
 - Why monitoring matters
 - Why organizations migrate to cloud
-

GitHub Tasks

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Week-4-Exam/

Files

exam-strategy.md

common-traps.md

interview-scenarios.md

By the End of This Session

You should become:

- more certification-ready,
 - more confident,
 - and more professionally articulate.
-

◆ SESSION 12 — SATURDAY

Final Mock Exam + Portfolio Completion

You Will Do

- Mock exam walkthrough
 - Weak-area correction
 - Final architecture review
 - Governance recap
 - Migration recap
 - Career guidance
-

◆ FINAL GITHUB SUBMISSION

Finalize:

README.md

README Must Include

- Professional summary
 - AWS services learned
 - Security concepts
 - Migration concepts
 - Architecture understanding
 - Screenshots
 - Certification target
 - Future roadmap
-

◎ FINAL STUDENT OUTCOMES

By the end of this bootcamp, you will have:

◆ Technical Understanding

- Cloud foundations
- Security awareness
- Compute understanding
- Storage strategy
- Networking basics
- Monitoring awareness
- Migration fundamentals
- Architecture reasoning

◆ Professional Assets

- GitHub portfolio
 - Structured documentation
 - Technical screenshots
 - Architecture notes
 - Scenario analysis evidence
-

◆ Career Readiness

You will be better prepared for:

- AWS Certified Cloud Practitioner
 - Cloud interviews
 - Entry-level cloud operations
 - Governance-aware cloud environments
-

● WHAT MAKES THIS PROGRAM DIFFERENT

This bootcamp does not produce:

students who only watched videos.

It produces:

- documented learners,
- governance-aware operators,
- cloud thinkers,
- and architecture-minded professionals.

Every student leaves with:

- training,
- practical exposure,
- visible proof of work,
- and structured cloud reasoning capability.